



AI Meets Academia: Enhancing Statistics Education with ChatGPT in Undergraduate Courses

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Abstract

This study explores the integration of ChatGPT as a learning tool in a second-year mathematical statistics course, comparing its effectiveness to traditional recorded lectures. Two challenging topics, order statistics and moment generating functions, were chosen to assess student learning outcomes. A mixed-methods approach was employed, involving quiz performance and survey feedback. Quantitative analysis revealed no significant difference in quiz scores between the ChatGPT group and the traditional learning group (Mann–Whitney test: p -value = 0.98), with average scores of 35.48% and 58.39% for Quiz 1 and Quiz 2 respectively in the ChatGPT group, compared to 34.86% and 58.25% in the traditional group. Qualitative feedback highlighted high satisfaction with ChatGPT's explanations (78.5% satisfaction with instructions and 81.3% overall experience rating). However, concerns about reliability (68.8%) and academic integrity (18.8%) were noted. Traditional lectures were valued for clarity (85.1% satisfaction) and overall experience (88.8% rated positively). This research underscores ChatGPT's potential to complement traditional teaching methods, offering flexibility and personalized support while addressing challenges in AI-assisted education. The findings inform future strategies for integrating AI into higher education, enhancing accessibility, engagement, and learning outcomes.

Keywords Artificial Intelligence · ChatGPT · Statistics Education · Academic Performance · Ethical AI · Student Engagement · Hybrid Learning Models

Introduction

The integration of Artificial Intelligence (AI) into education has revolutionized traditional learning environments, introducing tools that enhance accessibility, engagement,

and personalized instruction. Among these tools, ChatGPT, developed by OpenAI, has emerged as a standout innovation, capable of providing interactive and adaptive learning experiences (Dempere et al., 2023). Unlike conventional educational technologies, ChatGPT engages dynamically with students, addressing individual learning needs in real time. This adaptability has sparked growing interest in the potential of AI-driven tools to complement or even transform traditional instructional methods, particularly in disciplines like statistics that require intensive conceptual understanding (Ayeni et al., 2024).

Despite growing enthusiasm for AI in education, research into its application in technical fields such as mathematical statistics remains limited. Existing studies predominantly examine broad applications of AI tools, such as automating administrative tasks, providing basic tutoring, or supporting general learning activities. However, these studies often overlook specialized fields that require deep conceptual understanding and advanced computational skills. This study addresses this gap by exploring ChatGPT's potential as a supplementary learning

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tool in statistics education. Specifically, it evaluates its impact on two challenging topics: Order Statistics and Moment Generating Functions. These topics are not only foundational to mathematical statistics but also known for their conceptual difficulty among undergraduate students (Dhananjaya et al., 2024; Ellis & Slade, 2023).

AI-powered tools like ChatGPT have demonstrated their ability to support students by offering on-demand explanations, answering complex questions, and tailoring resources to individual needs. These capabilities have been linked to enhanced student engagement, improved comprehension of challenging concepts, and the development of critical thinking and problem-solving skills (Lo, 2023; Wahba et al., 2024). Such advantages are particularly pronounced in statistics, where students frequently struggle with abstract concepts such as order statistics and moment generating functions. Studies suggest that AI can serve as a valuable adjunct to traditional learning methods, providing an additional layer of support that can improve student performance and satisfaction (Dhananjaya et al., 2024; Ellis & Slade, 2023). However, the integration of AI tools into education also raises significant challenges. Concerns about academic integrity, the accuracy of AI-generated content, and the potential for overreliance of students on AI solutions are barriers to widespread adoption (Adeshola & Adepoju, 2023). Ethical issues, such as data privacy and algorithmic bias, further complicate the landscape of AI in education (Neumann et al., 2023). These challenges underscore the need for evidence-based research to evaluate the impact of AI tools on educational outcomes and to develop strategies for their responsible implementation.

This study contributes to the ongoing discourse by empirically assessing the effectiveness of ChatGPT as a supplemental learning tool in comparison to traditional pre-recorded lectures. Through a mixed-methods approach, it examines students' academic performance and perceptions of ChatGPT's utility, academic integrity, and overall satisfaction. By focusing on two technically demanding topics in statistics, this research provides insights into how AI-driven tools can address specific educational challenges.

The paper is organized as follows. Section 2 reviews relevant literature on AI in education and its applications in technical fields. Section 3 outlines the research methodology, including details of the experimental design, population sample, and survey instrument. Section 4 presents an analysis of quiz performance and survey feedback. Section 5 discusses the implications of the findings for educators and policymakers. Finally, Section 6 concludes with recommendations for future research and the integration of AI tools in higher education.

Literature Review

The rapid proliferation of artificial intelligence (AI) in education over the past decade has been remarkable. Talan (2021) highlights exponential growth in AI-related publications, with education emerging as the most studied field in terms of article volume and citation impact. Among AI tools, ChatGPT, developed by OpenAI, has gained significant attention for its ability to enhance learning experiences through interactive and on-demand assistance.

Research highlights the potential of AI to address critical gaps in education. Lo (2023) discusses how ChatGPT can create customized learning experiences, offering personalized support to students from diverse backgrounds and enabling them to overcome barriers to learning. In particular, Lo emphasizes the AI's capacity to provide differentiated instruction, catering to varying learning speeds and styles. Dhananjaya et al. (2024) explore how ChatGPT can enhance student engagement by automating tasks such as grading and providing immediate feedback on routine queries, thereby allowing instructors to focus on more complex aspects of teaching. Winkler and Söllner (2018) expand on these findings, showing how conversational AI tools like chatbots have transformed student interactions, creating new opportunities for active learning.

Ellis and Slade (2023) demonstrate that ChatGPT is especially effective in technical fields like statistics and data science, where students often struggle with complex concepts and coding exercises. Their research shows that ChatGPT helps students not only understand theoretical material but also engage in hands-on activities like coding, improving their overall learning experience. Similarly, Wahba et al. (2024) find that incorporating ChatGPT into undergraduate statistics courses significantly improved students' statistical reasoning and positively influenced their attitudes toward quantitative subjects. Fauzi et al. (2023) further support this, showing that ChatGPT increases student productivity by offering quick access to information, enhancing language skills, and streamlining collaboration. Holmes et al. (2019) emphasize the broader promise of AI in personalized learning, highlighting its ability to adapt to individual students' needs and provide actionable insights for instructors.

While AI offers significant potential and a broad range of applications in higher education, several challenges persist. Adeshola and Adepoju (2023) warn that students' overreliance on AI-generated content could promote academic dishonesty, a concern that remains significant given the ease with which students can generate plausible but potentially misleading content. Ethical issues, including concerns about data privacy and AI biases, also need to be addressed (Neumann et al., 2023).

Selwyn (2019) critiques the implications of AI's increasing role in education, particularly the risks of eroding traditional pedagogical relationships and creating a dependence on technological systems. Moreover, while AI detection tools exist, they are not always reliable in identifying AI-generated content, further complicating efforts to uphold academic integrity. Kooli (2023) confirms this concern, suggesting that although ChatGPT can produce high-quality responses, its misuse could lead to serious ethical dilemmas in academia.

The use of AI in technical fields is underexplored. Binns (2018) highlights how biases in machine learning algorithms can perpetuate inequities in educational outcomes, urging developers to prioritize fairness and transparency in AI models. Despite these concerns, the potential of AI tools like ChatGPT to support students in technical subjects such as statistics is increasingly recognized. Wahba et al. (2024) highlight how ChatGPT enhances students' statistical reasoning skills and fosters positive attitudes towards quantitative learning. Zawacki-Richter et al. (2019) provide a systematic review of AI applications in higher education, offering insights into how educators can effectively integrate these tools to complement traditional methods. This study builds on such insights by exploring ChatGPT's utility in addressing conceptual challenges in statistics education.

While prior studies underscore the promise of AI in improving student engagement, personalized learning, and productivity, there remains a lack of empirical research focusing on its application in conceptually intensive subjects like statistics. Recent comparative studies across other fields, such as engineering, language learning, and medical education, demonstrate the varied success of AI-assisted instruction (Ausat et al., 2023; Dogan et al., 2023). However, these studies often overlook domain-specific rigor and fail to evaluate cognitive depth in learning outcomes. This research extends the existing literature by empirically comparing the academic performance of students using ChatGPT with those using traditional recorded lectures, focusing on measurable learning outcomes rather than engagement alone. By addressing both benefits and challenges, this study contributes to a more nuanced understanding of ChatGPT's role in education. In doing so, it aligns with constructivist perspectives that emphasize learner-centered, interactive environments conducive to deeper understanding.

Across the literature, two key tensions emerge: first, the promise of AI for personalization and engagement versus concerns about reliability and academic dishonesty; second, enthusiasm for AI in general education versus a lack of empirical work in technically rigorous domains. This study addresses both gaps by exploring how ChatGPT performs not only in terms of learning outcomes, but also in shaping student experiences in a conceptually demanding statistics course.

Research Methodology

This study employs a mixed-methods approach, integrating quantitative and qualitative analyses to evaluate the effectiveness of ChatGPT as a learning tool compared to traditional recorded lectures. Two foundational topics in mathematical statistics, Order Statistics and Moment Generating Functions, were selected due to their conceptual difficulty and importance in the course curriculum. The quantitative analysis uses the Mann–Whitney test, a non-parametric method robust to violations of normality, to compare quiz scores between the ChatGPT and traditional lecture groups (Mann & Whitney, 1947). Additionally, the Shapiro–Wilk test is applied to assess the normality of data distributions (Shapiro & Wilk, 1965).

To complement the quantitative results, qualitative data were collected through a structured survey, which captured students' perceptions, satisfaction, and concerns about the learning tools. The survey responses were analyzed using thematic analysis to extract patterns and insights (Braun & Clarke, 2006), offering a deeper understanding of student experiences and attitudes.

Experimental Design

This study evaluates the effectiveness of ChatGPT as a learning tool compared to traditional recorded lectures in a second-year mathematical statistics course. Students were randomly assigned to one of two groups: the ChatGPT group (experimental) or the recorded lecture group (control). The randomized design supports stronger causal inferences between instructional method and student outcomes (Montgomery, 2019). Each quiz consisted of one question with three parts, assessing both conceptual understanding and computational skills. Quizzes were administered in person, limited to 25 min, and required handwritten solutions without the use of calculators or AI tools. The ChatGPT group used structured prompts to explore concepts through text-based interaction. A copy of these prompts is available upon request from the corresponding author. The recorded lecture group viewed 30–40 min pre-recorded sessions without interactive elements. No statistical files were uploaded, and all ChatGPT interactions were anonymous and prompt-driven. This design ensured a balanced and controlled comparison between AI-assisted and traditional instructional approaches.

Both groups studied two challenging topics, Order Statistics and Moment Generating Functions, which are integral to the course curriculum. The ChatGPT group was provided with guided instructions on effectively using the tool, including prompts for exploring theoretical concepts,

solving problems, and generating additional resources. Meanwhile, the traditional recording group accessed pre-recorded lectures created by the course instructor, ensuring consistency and rigor in content delivery.

The study was conducted over a two-week period:

- **Week 1:** Students engaged with materials on Order Statistics, completing assigned practice problems and an in-person quiz at the end of the week.
- **Week 2:** The same structure was followed for Moment Generating Functions.

Both groups had access to identical practice problems, final answers for self-assessment, and scheduled office hours for additional support. Quizzes were designed to assess understanding, problem-solving skills, and the ability to apply concepts, ensuring a fair and uniform evaluation of both groups.

Population and Sample

The participants for this study were undergraduate students enrolled in a second-year mathematical statistics course at a major Canadian university. Out of 97 students registered in the course, 70 volunteered to participate after an open call for participation. Approximately 95% of the participants were pursuing minor, major, or specialist programs in statistics, while the remaining students were from related disciplines such as mathematics or computer science.

Participants were randomly assigned to one of the two groups, ensuring balanced demographic and academic characteristics across both groups. Out of 70 students who wrote the quizzes, 62 responded to the questionnaire after the experiment. The majority (80%) were third-year students or higher, indicating substantial prior exposure to statistical coursework and a solid statistical foundation.

Table 1 summarizes the demographic breakdown, including gender, year of study, and academic performance in the prerequisite course. This diversity provided a robust basis for evaluating the effectiveness of ChatGPT as a learning tool.

Survey Instrument

Qualitative feedback was gathered using a structured questionnaire designed to evaluate students' experiences with the two learning tools, ChatGPT and traditional recorded lectures. The questionnaire addressed several key areas, including demographic information, time spent on study preparation, satisfaction levels with the learning tools and resources, and specific concerns or challenges encountered during the study. The questionnaire was designed to address several key areas:

Table 1 The demographic information about the volunteers

Covariate	Total (<i>n</i> = 62)
Gender	
Male	49 (79.03%)
Female	12 (19.35%)
Others	1 (1.61%)
Program of Study	
Specialist in statistic	7 (11.29%)
Major in statistic	36 (58.06%)
Minor in statistic	15 (24.19%)
CS or Math	4 (6.45%)
Year of Study	
Second year	12 (19.35%)
Third year	35 (56.45%)
Fourth year	12 (19.35%)
Fifth year or more	3 (4.84%)
Final Grades in Prerequisite Course	
50–59	15 (24.19%)
60–69	17 (27.42%)
70–79	14 (22.58%)
80–89	11 (17.74%)
90–100	5 (8.06%)

- **Demographic Information:** Gender, program of study, year of study, and performance in the relevant prerequisite course.
- **Study Time:** Time spent preparing for each quiz, categorized into intervals (e.g., 1–2 h, 2–3 h).
- **Satisfaction Levels:** Students rated their satisfaction with the provided instructions, practice problems, and overall experience using a Likert scale ranging from “extremely satisfied” to “not satisfied at all.”
- **Concerns and Perceptions:** Identifying factors that might have prevented students from using ChatGPT as a supplementary tool, such as concerns about reliability, academic integrity, or personal learning preferences.
- **Features of Learning Tools:** Questions asked students to identify the most helpful features of ChatGPT or traditional recordings, such as clarity of explanations, problem-solving assistance, or interactivity.

The questionnaire was reviewed and refined by three professors and two senior undergraduate students from the department to ensure clarity, validity, and alignment with the study's objectives. It was administered in English and designed to maintain participant anonymity, encouraging honest and unbiased responses. This structured approach ensured the survey captured diverse perspectives, enabling a comprehensive analysis of the learning tools. The full questionnaire is available from the corresponding author upon request.

Analysis of Quiz Performance and Survey Feedback

This section evaluates the effectiveness of ChatGPT as a learning tool through an analysis of quiz performance and survey feedback. The quantitative analysis compares quiz scores between the ChatGPT and traditional recording groups, focusing on metrics such as understanding, problem-solving ability, and the application of concepts. In parallel, the qualitative analysis examines students' experiences, satisfaction levels, and concerns, offering a deeper understanding of their engagement with the two learning methods. Together, these insights provide a comprehensive assessment of the advantages and limitations of ChatGPT in comparison to the traditional instructional methods.

Quiz Data Analysis

The quiz results were analyzed to evaluate the effectiveness of ChatGPT compared to traditional recorded lectures in supporting student learning. Two quizzes were administered, corresponding to the two topics covered in the study: Order Statistics (Quiz 1) and Moment Generating Functions (Quiz 2). Both quizzes were identical for the two groups and designed to assess understanding, problem-solving skills, and the ability to apply concepts. Regarding quiz participation, for Quiz 1, the ChatGPT group had 35 participants, the Recording group had 31. For Quiz 2, the ChatGPT group had 40 participants, and the Recording group had 31 participants.

The average scores for the ChatGPT group were 35.48% for Quiz 1 and 58.39% for Quiz 2, while the traditional recording group scored averages of 34.86% for Quiz 1 and 58.25% for Quiz 2. The median scores were also similar across both groups, indicating comparable performance levels.

Statistical tests were conducted to validate these observations. Since the data did not follow a normal distribution, as determined by the Shapiro–Wilk test (p -value = 0.003), the non-parametric Mann–Whitney test was used to compare the quiz scores. The results of the Mann–Whitney test showed no significant difference between the two groups for Quiz 1 (p -value = 0.985) or Quiz 2 (p -value = 0.986), suggesting that both learning tools were equally effective in supporting student performance.

These findings indicate that ChatGPT is a viable alternative to traditional recorded lectures, providing comparable outcomes in understanding and applying mathematical statistics concepts. The similar performance across groups suggests that ChatGPT can offer the same educational value as traditional methods, with the added benefit of interactivity and accessibility.

Survey Feedback and Observations

The survey offered valuable insights into students' experiences with ChatGPT and traditional recorded lectures, capturing their satisfaction levels, preferences, and key concerns regarding these two learning tools. This analysis incorporates both quantitative and qualitative feedback to assess the strengths and limitations of each approach comprehensively.

Of the 70 students who completed the quizzes, 62 responded to the survey. Among them, 34 students provided feedback on their experience with ChatGPT, while 28 students shared their views on recorded lectures. These responses form the basis for evaluating the comparative effectiveness and student reception of the two methods.

To better understand student satisfaction, preferences, and challenges, the survey findings were categorized into the following key themes:

Satisfaction Levels

Participants reported high levels of satisfaction with both tools, with slightly higher overall satisfaction observed for traditional recorded lectures. Among participants in the ChatGPT group:

- 78.5% expressed satisfaction or extreme satisfaction with the instructions for using the tool.
- 84.4% were satisfied with the practice problems provided.
- 81.3% rated their overall experience positively.

In comparison, participants in the traditional recording group reported:

- 85.1% satisfaction with the practice materials.
- 88.8% rated their overall experience positively.

Helpful Features

Students identified specific features of each tool that they found most beneficial:

- **ChatGPT:** Participants appreciated its interactivity, tailored explanations, step-by-step problem-solving assistance, and instant access to resources.
- **Traditional Recorded Lectures:** Participants valued the clear, structured delivery of material and the ability to revisit sections for clarification.

Concerns and Challenges

Despite positive feedback, participants highlighted several challenges with ChatGPT:

- **Reliability and Accuracy:** 68.8% of participants expressed concerns about the accuracy of ChatGPT's responses, noting that it sometimes provided incomplete or incorrect explanations, requiring additional verification.
- **Academic Integrity:** 18.8% of participants raised concerns about potential misuse of ChatGPT for academic dishonesty, emphasizing the need for ethical guidelines on AI use.

For the traditional recording group, the primary concern was the lack of interactivity and opportunities for immediate clarification, which some participants found limiting for complex topics.

Thematic Insights

Open-ended survey responses offered additional observations:

- Participants in the ChatGPT group highlighted the tool's accessibility and flexibility but suggested that more detailed guidance or structured prompts could enhance its effectiveness.
- Participants in the traditional recording group valued the familiarity and consistency of lectures but noted that dynamic features, such as interactive elements, could improve engagement.

Overall Observations

The survey results reveal complementary strengths of the two learning tools. ChatGPT excels in interactivity and accessibility but faces challenges related to reliability and ethical considerations. In contrast, traditional recorded lectures provide structure and clarity but lack the responsiveness that many students find beneficial in AI-driven tools. These findings suggest that a hybrid approach, combining the strengths of both methods, could further enhance the learning experience by leveraging ChatGPT for interactive exploration and recorded lectures for foundational understanding.

Discussion and Implications

The findings of this study underscore the potential of ChatGPT as a supplementary learning tool in higher education, particularly in statistics education. The lack of significant differences in quiz performance between the ChatGPT and traditional lecture groups demonstrates that AI-based tools can effectively complement traditional learning methods. This parity in outcomes suggests that ChatGPT can offer comparable educational value while providing additional benefits such as interactivity and accessibility.

ChatGPT's adaptability to individual learning needs positions it as a valuable resource for addressing diverse learner requirements. Its ability to provide instant feedback, tailored explanations, and interactive problem-solving aligns with principles of active learning and constructivist pedagogy (Holmes et al., 2019). These features enable students to explore complex topics like order statistics and moment-generating functions independently, bridging gaps in understanding that traditional lectures might leave unaddressed.

The findings also support the integration of ChatGPT into hybrid or flipped classroom models. By leveraging ChatGPT for pre-class preparation, instructors can reserve in-person sessions for collaborative activities and higher-order cognitive tasks. This approach aligns with Bloom's revised taxonomy, where foundational knowledge is developed outside the classroom, allowing educators to focus on critical thinking and problem-solving during class discussions (Wahba et al., 2024).

Despite its potential, ChatGPT's integration into education is not without challenges. While quiz performance between the ChatGPT and traditional groups was statistically equivalent, survey data revealed experiential distinctions. Students using ChatGPT appreciated the interactivity and autonomy, while others preferred the structured clarity of recorded lectures. This suggests that parity in outcomes does not equate to pedagogical equivalence, a key insight for instructional design. Furthermore, ethical considerations are critical. Concerns about the reliability of AI-generated responses, risks of academic dishonesty, and institutional responsibility for guiding AI use must be addressed. Recent work by Adeshola and Adepoju (2023), Kooli (2023), and Neumann et al. (2023) reinforces the urgency of developing guidelines for accuracy, transparency, and ethical use in AI-supported education.

Concerns about the accuracy of AI-generated content and the risk of overreliance highlight the need for careful implementation. Educators must play an active role in curating AI-generated materials to ensure alignment with curricular objectives and to mitigate errors. Incorporating verification mechanisms and user training can further enhance the tool's reliability.

Academic integrity remains a critical issue. The ease with which ChatGPT can generate plausible responses raises concerns about misuse, necessitating the development of clear ethical guidelines for its use in education. Assessments that prioritize application and analysis over rote memorization can discourage academic dishonesty and encourage students to critically engage with AI-generated outputs.

ChatGPT offers scalable solutions to challenges faced in large or resource-constrained classrooms, where personalized instruction is often limited. By enabling students to engage with material on their own terms, ChatGPT addresses educational inequities, providing consistent support to learners from diverse backgrounds. This aligns with recommendations in Zawacki-Richter et al. (2019), advocating for AI's role in reducing disparities in access to quality education.

However, issues of digital equity must also be addressed. Ensuring that all students have access to the necessary technology and training to use AI tools effectively is essential for their equitable integration into education. Institutions must prioritize investments in infrastructure and professional development to support both educators and students in adopting AI-driven learning methods.

The study's findings open several avenues for future research. Longitudinal studies are needed to evaluate the long-term impact of ChatGPT on learning outcomes, knowledge retention, and the development of higher-order thinking skills. Additionally, exploring ChatGPT's application across different disciplines and educational levels can provide broader insights into its utility and limitations. Further research should also examine the integration of ChatGPT into collaborative learning environments. While this study focused on individual learning outcomes, understanding how AI tools can enhance group dynamics and peer-based learning could inform strategies for their use in diverse instructional settings.

This study demonstrates that ChatGPT is a viable alternative to traditional recorded lectures, offering comparable outcomes in understanding and applying mathematical statistics concepts. Its strengths in interactivity, accessibility, and adaptability highlight its potential to enhance learning experiences in higher education. However, addressing challenges related to reliability, academic integrity, and digital equity is critical for maximizing its impact while minimizing its drawbacks.

By leveraging the complementary strengths of AI tools and traditional methods, educators can create more inclusive, effective, and engaging learning environments. The findings of this study underscore the need for ongoing research and collaboration between educators and AI developers to ensure that tools like ChatGPT are integrated responsibly and equitably into education.

Limitations

While this study highlights the potential of ChatGPT as a supplementary learning tool in higher education, particularly in the domain of statistics, several limitations must be acknowledged. These limitations provide context for interpreting the findings and guide areas for future research.

1. **Sample Size and Demographics:** The study was conducted with a relatively small sample size of students from a single course and institution. This limits the generalizability of the findings to broader educational contexts or diverse demographic groups. Future studies should aim to include larger and more varied populations to enhance the applicability of the results.
2. **Short-Term Assessment:** This study primarily evaluated short-term academic performance through quiz scores. Long-term effects, such as knowledge retention, development of higher-order thinking skills, and impact on collaborative learning, remain unexplored. Longitudinal studies are necessary to understand the sustained influence of ChatGPT on learning outcomes.
3. **Limited Scope of Topics:** The research is focused on two challenging topics within statistics: order statistics and moment-generating functions. While these are representative of advanced statistical concepts, the findings may not fully capture ChatGPT's efficacy across other topics or disciplines. Future research should investigate its application across a broader range of subjects.
4. **Reliability of AI-Generated Content:** Concerns about the accuracy and reliability of ChatGPT-generated content were raised by participants. These issues underscore the need for active oversight by educators to ensure that AI tools deliver accurate and curriculum-aligned materials. Incorporating verification mechanisms and user training could address these concerns.

Ethical Concerns:

5. The potential for academic dishonesty, such as overreliance on AI-generated solutions, presents a significant challenge. Designing assessments that prioritize critical thinking and originality is essential to mitigate this risk. Institutions must develop clear guidelines for the ethical use of AI in educational settings.
6. **Limited Feedback on Collaborative Learning:** While ChatGPT offers personalized support, its role in fostering collaborative and peer-based learning environments remains unclear. Future studies could explore how Chat-

GPT integrates with group learning dynamics and its impact on student interactions.

By acknowledging these limitations, the study highlights areas for further investigation and improvement. Addressing these challenges will be critical for leveraging ChatGPT effectively as a tool to enhance learning outcomes and address educational disparities.

Conclusion

This study demonstrates that ChatGPT is a viable alternative to traditional recorded lectures, offering comparable outcomes in understanding and applying mathematical statistics concepts. The findings show that both ChatGPT and traditional lectures can effectively support student learning, with ChatGPT providing additional benefits such as interactivity, instant feedback, and accessibility. These features position ChatGPT as a valuable tool for addressing diverse learner needs and enhancing engagement in higher education.

The study's results also highlight practical implications for integrating ChatGPT into educational settings. By leveraging ChatGPT in flipped classroom models (Lage et al., 2000) or hybrid learning environments, educators can facilitate pre-class preparation and reserve in-person sessions for collaborative and higher-order learning activities. It is important to note that while “hybrid learning” and “blended learning” are often used interchangeably, they have nuanced differences: hybrid learning typically combines in-person and online instruction with flexibility for students to choose between the two modes, whereas blended learning focuses on a structured integration of online and face-to-face activities (Hrastinski, 2019). This approach aligns with insights from Bonk and Graham (2006), emphasizing the transformative potential of blended learning models in optimizing instructional design. Educators could implement these models by providing students with targeted prompts for pre-class exploration using ChatGPT, followed by in-class discussions and group problem-solving activities that reinforce key concepts. This approach aligns with modern pedagogical frameworks, enabling more effective use of classroom time and supporting active learning.

However, challenges related to the reliability of AI-generated content, academic integrity, and digital equity must be addressed. Ensuring access to necessary technology, providing proper training, and developing ethical guidelines for AI use are essential steps toward the responsible integration of tools like ChatGPT. Educators must remain active curators of AI-generated materials, ensuring alignment with curricular goals and mitigating potential errors.

Future research should explore the long-term impact of ChatGPT on learning outcomes, including knowledge

retention, higher-order thinking skills, and collaborative learning dynamics. Additionally, expanding the scope to include diverse disciplines and student populations will provide broader insights into the tool's utility and limitations. For instance, exploring ChatGPT's effectiveness in fields like engineering or humanities could offer a deeper understanding of its interdisciplinary applications. Investigating strategies for integrating ChatGPT into group-based learning environments could further enhance its role in education.

In conclusion, ChatGPT has the potential to complement traditional instructional methods, creating more inclusive, effective, and engaging learning environments. A hybrid approach, where AI tools support exploration and traditional methods offer structure, may yield optimal outcomes. The study's design and instructional prompts offer a model for replication in other disciplines, including engineering, computer science, and the humanities. Ongoing collaboration between educators, researchers, and AI developers is essential for ensuring responsible integration and sustained pedagogical impact.

Data Availability Statement

The data that support the findings of this study are not available due to ethical restrictions.

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Declarations

Ethical Statement In accordance with the comprehensive ethics standards at a university in Canada, this authorization has been granted by the university's Ethics Review Committee.

Conflict of Interests The authors declare no conflict of interest.

Competing Interests All authors declare that they have no competing financial or non-financial interests directly or indirectly related to the work submitted for publication.

References

- Adeshola, I., & Adepoju, A. P. (2023). The opportunities and challenges of ChatGPT in education. *Interactive Learning Environments*, 32(10), 6159–6172. <https://doi.org/10.1080/10494820.2023.2253858>
- Ausat, A., Massang, B., Efendi, M., Nofirman, N., & Riady, Y. (2023). Can Chat GPT replace the role of the teacher in the classroom: A fundamental analysis. *Journal on Education*, 5, 16100–16106. <https://doi.org/10.31004/joe.v5i4.2745>

- Ayeni, O. O., Al Hamad, N. M., Chisom, O. N., Osawaru, B., & Adewusi, O. E. (2024). AI in education: A review of personalized learning and educational technology. *GSC Advanced Research and Reviews*, 18(2), 261–271. <https://doi.org/10.30574/gscarr.2024.18.2.0062>
- Binns, R. (2018). Fairness in machine learning: Lessons from political philosophy. Proceedings of the 2018 Conference on Fairness, Accountability, and Transparency. <https://proceedings.mlr.press/v81/binns18a/binns18a.pdf>
- Bonk, C. J., & Graham, C. R. (2006). *The handbook of blended learning: Global perspectives, local designs*. John Wiley & Sons.
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101. <https://doi.org/10.1191/1478088706qp0630a>
- Dempere, J., Modugu, K., Hesham, A., & Ramasamy, L. K. (2023). The impact of ChatGPT on higher education. *Frontiers in Education*, 8, 1206936. <https://doi.org/10.3389/educ.2023.1206936>
- Dhananjaya, G. M., Goudar, R. H., Govindaraja, K., Kaliwal, R. B., Rathod, V. K., Deshpande, S. L., Kulkarni, A., & Hukkeri, G. S. (2024). Enhancing education with ChatGPT: Revolutionizing personalized learning and teacher support. *EAI Endorsed Transactions on Internet of Things*, 10, 1–9. <https://doi.org/10.4108/eetiot.6998>
- Dogan, M. E., Goru Dogan, T., & Bozkurt, A. (2023). The use of artificial intelligence (AI) in online learning and distance education processes: A systematic review of empirical studies. *Applied Sciences*, 13, 3056. <https://doi.org/10.3390/app13053056>
- Ellis, A. R., & Slade, E. (2023). A new era of learning: Considerations for ChatGPT as a tool to enhance statistics and data science education. *Journal of Statistics and Data Science Education*, 31(2), 128–133. <https://doi.org/10.1080/26939169.2023.2223609>
- Fauzi, F., Tuhuteru, L., Sampe, F., Ausat, A., & Hatta, H. (2023). Analysing the role of ChatGPT in improving student productivity in higher education. *Journal on Education*, 5(4), 14886–1491. <https://doi.org/10.31004/joe.v5i4.2563>
- Holmes, W., Bialik, M., & Fadel, C. (2019). *Artificial intelligence in education: Promises and implications for teaching and learning*. Center for Curriculum Redesign.
- Hrastinski, S. (2019). What do we mean by blended learning? *TechTrends*, 63(5), 564–569. <https://doi.org/10.1007/s11528-019-00375-5>
- Kooli, C. (2023). Chatbots in education and research: A critical examination of ethical implications and solutions. *Sustainability*, 15(7), 5614. <https://doi.org/10.3390/su15075614>
- Lage, M. J., Platt, G. J., & Treglia, M. (2000). Inverting the classroom: A gateway to creating an inclusive learning environment. *The Journal of Economic Education*, 31(1), 30–43. <https://doi.org/10.2307/1183338>
- Lo, C. K. (2023). What is the impact of ChatGPT on education? A rapid review of the literature. *Education Sciences*, 13(4), 410. <https://doi.org/10.3390/educsci13040410>
- Mann, H. B., & Whitney, D. R. (1947). On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics*, 18(1), 50–60. <https://doi.org/10.1214/aoms/1177730491>
- Montgomery, D. C. (2019). *Design and analysis of experiments* (10th ed.). John Wiley & Sons.
- Neumann, M., Rauschenberger, M., & Schön, E.-M. (2023). We need to talk about ChatGPT: The future of AI and higher education. In 2023 IEEE/ACM 5th International Workshop on Software Engineering Education for the Next Generation (SEENG) (29–32). <https://doi.org/10.1109/seeng59157.2023.00010>
- Selwyn, N. (2019). *Should robots replace teachers? AI and the future of education*. Polity Press.
- Shapiro, S. S., & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika*, 52(3–4), 591–611. <https://doi.org/10.1093/biomet/52.3-4.591>
- Talan, T. (2021). Artificial intelligence in education: A bibliometric study. *International Journal of Research in Education and Science*, 7(3), 822–837. <https://doi.org/10.46328/ijres.2409>
- Wahba, F., Ajlouni, A. O., & Abumosa, M. A. (2024). The impact of ChatGPT-based learning statistics on undergraduates' statistical reasoning and attitudes toward statistics. *Eurasia Journal of Mathematics, Science and Technology Education*, 20(7), em2468. <https://doi.org/10.29333/ejmste/14726>
- Winkler, R., & Söllner, M. (2018). Unleashing the potential of chatbots in education: A state-of-the-art analysis. *Academy of Management Proceedings*. <https://api.semanticscholar.org/CorpusID:186258485>
- Zawacki-Richter, O., Marín, V. I., Bond, M., & Gouverneur, F. (2019). Systematic review of research on artificial intelligence applications in higher education: Where are the educators? *International Journal of Educational Technology in Higher Education*, 16, 39. <https://doi.org/10.1186/s41239-019-0171-0c>

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